

Chapter 3. Determinants

Sim, Min Kyu, Ph.D., mksim@seoultech.ac.kr



1 3.1. Introduction to Determinants

2 3.2. Properties of Determinants

3 3.3. Cramer's Rule, Volume, and Linear Transformations

3.1. Introduction to Determinants

Definition

- For $n \geq 2$, the **determinant** of an $n \times n$ matrix $A = [a_{ij}]$ is the sum of n terms of the form $\pm a_{1i} \det A_{1i}$, with plus and minus signs alternating, where the entries $a_{11}, a_{12}, \dots, a_{1n}$ are from the first row of A . In symbols,

$$\begin{aligned} \det A &= a_{11} \det A_{11} - a_{12} \det A_{12} + \cdots + (-1)^{1+n} a_{1n} \det A_{1n} \\ &= \sum_{j=1}^n (-1)^{1+j} a_{1j} \det(A_{1j}) \end{aligned}$$

- **Example 1.** Compute the determinant of

$$A = \begin{bmatrix} 1 & 5 & 0 \\ 2 & 4 & -1 \\ 0 & -2 & 0 \end{bmatrix}$$

- **Solution:** Compute

$$\begin{aligned} \det A &= a_{11} \det A_{11} - a_{12} \det A_{12} + a_{13} \det A_{13} \\ &= 1 \cdot \det \begin{bmatrix} 4 & -1 \\ -2 & 0 \end{bmatrix} - 5 \cdot \det \begin{bmatrix} 2 & -1 \\ 0 & 0 \end{bmatrix} + 0 \cdot \det \begin{bmatrix} 2 & 4 \\ 0 & -2 \end{bmatrix} \\ &= 1(0 - 2) - 5(0 - 0) + 0(-4 - 0) = -2 \end{aligned}$$

- Another common notation for the determinant of a matrix uses a pair of vertical lines in place of brackets, i.e. $\det A = |A|$
- Thus the calculation in Example 1 can be written as

$$\det A = 1 \cdot \begin{vmatrix} 4 & -1 \\ -2 & 0 \end{vmatrix} - 5 \cdot \begin{vmatrix} 2 & -1 \\ 0 & 0 \end{vmatrix} + 0 \cdot \begin{vmatrix} 2 & 4 \\ 0 & -2 \end{vmatrix} = \dots = -2$$

- To state the next theorem, it is convenient to write the definition of $\det A$ in a slightly different form. Given $A = [a_{ij}]$, the (i, j) —**cofactor** of A is the number C_{ij} given by

$$C_{ij} = (-1)^{i+j} \det A_{ij} \quad (4)$$

- Then

$$\det A = a_{11}C_{11} + a_{12}C_{12} + \cdots + a_{1n}C_{1n}$$

- This formula is called a **cofactor expansion across the first row** of A .
- **Theorem 1:** The determinant of an $n \times n$ matrix A can be computed by a cofactor across any row or down any column.
 - The expansion across the i -th row using the cofactors in (4) is

$$\det A = a_{i1}C_{i1} + a_{i2}C_{i2} + \cdots + a_{in}C_{in}$$

- The cofactor expansion down the j th column is

$$\det A = a_{1j}C_{1j} + a_{2j}C_{2j} + \cdots + a_{nj}C_{nj}$$

- **Example 2.** Use a cofactor expansion across the third row to compute $\det A$, where

$$A = \begin{bmatrix} 1 & 5 & 0 \\ 2 & 4 & -1 \\ 0 & -2 & 0 \end{bmatrix}$$

- **Solution:** Compute

$$\begin{aligned} \det A &= a_{31}C_{31} + a_{32}C_{32} + a_{33}C_{33} \\ &= (-1)^{3+1}a_{31}\det A_{31} + (-1)^{3+2}a_{32}\det A_{32} + (-1)^{3+3}a_{33}\det A_{33} \\ &= 0 \cdot \begin{bmatrix} 5 & 0 \\ 4 & -1 \end{bmatrix} - (-2) \begin{bmatrix} 1 & 0 \\ 2 & -1 \end{bmatrix} + 0 \begin{bmatrix} 1 & 5 \\ 2 & 4 \end{bmatrix} \\ &= 0 + 2(-1) + 0 = -2 \end{aligned}$$

- **Theorem 2:** If A is a triangular matrix, then $\det A$ is the product of the entries on the main diagonal of A .

Suggested Exercises

- 3.1.4
- 3.1.10

3.2. Properties of Determinants

- The Theorem 3 below answers to a question “How does an elementary row operation affect determinant?”
- **Theorem 3:** Let A be a square matrix
 - a) (*Replacement*) If a multiple of one row of A is added to another row to produce a matrix B , then $\det B = \det A$.
 - b) (*Interchange*) If two rows of A are interchanged to produce B , then $\det B = -1 \cdot \det A$.
 - c) (*Scaling*) If one row of A is multiplied by k to produce B , then $\det B = k \cdot \det A$.

- **Example 1** Compute $\det A$, where $A = \begin{bmatrix} 1 & -4 & 2 \\ -2 & 8 & -9 \\ -1 & 7 & 0 \end{bmatrix}$

- **Solution:**

- The strategy is to reduce A to echelon form and then to use the fact that the determinant of a triangular matrix is the product of the diagonal entries. The first two row replacements in column 1 do not change the determinant:

$$\det A = \begin{vmatrix} 1 & -4 & 2 \\ -2 & 8 & -9 \\ -1 & 7 & 0 \end{vmatrix} = \begin{vmatrix} 1 & -4 & 2 \\ 0 & 0 & -5 \\ -1 & 7 & 0 \end{vmatrix} = \begin{vmatrix} 1 & -4 & 2 \\ 0 & 0 & -5 \\ 0 & 3 & 2 \end{vmatrix}$$

- An interchange of rows 2 and 3 reverses the sign of the determinant, so

$$\det A = - \begin{vmatrix} 1 & -4 & 2 \\ 0 & 3 & 2 \\ 0 & 0 & -5 \end{vmatrix} = -(1)(3)(-5) = 15$$

- **Theorem 4:** A square matrix A is invertible if and only if $\det A \neq 0$.

- **Example 3.** Compute $\det A$, where $A = \begin{bmatrix} 3 & -1 & 2 & 5 \\ 0 & 5 & -3 & -6 \\ -6 & 7 & -7 & 4 \\ -5 & -8 & 0 & 9 \end{bmatrix}$

- **Solution**

- Add 2 times row 1 to row 3 ($R3 \leftarrow R3 + 2R1$) to obtain

$$\det A = \det \begin{bmatrix} 3 & -1 & 2 & 5 \\ 0 & 5 & -3 & -6 \\ 0 & 5 & -3 & -6 \\ -5 & -8 & 0 & 9 \end{bmatrix} = 0$$

- because the second and third rows of the matrix are equal.

Column Operations

- **Theorem 5:** If A is a matrix, then $\det A^T = \det A$.
- **Proof**
 - The theorem is obvious for $n = 1$.
 - Suppose the theorem is true for $k \times k$ determinants and let $n = k + 1$. Then the cofactor of a_{1j} in A equals the cofactor of a_{j1} in A^T , because the cofactors involve $k \times k$ determinants. Hence the cofactor expansion of $\det A$ along the first row equals the cofactor expansion of $\det A^T$ down the first column. That is, A and A^T have equal determinants.
 - Thus the theorem is true for $n = 1$, and the truth of the theorem for one value of k implies its truth for the next value of $k + 1$. By the principle of *mathematical induction*, the theorem is true for all $n \geq 1$.

Determinants and Matrix Products

- **Theorem 6:** If A and B are matrices, then $\det AB = (\det A)(\det B)$

- **Example 5** Verify Theorem 6 for $A = \begin{bmatrix} 6 & 1 \\ 3 & 2 \end{bmatrix}$ and $B = \begin{bmatrix} 4 & 3 \\ 1 & 2 \end{bmatrix}$.

- **Solution**

•

$$AB = \begin{bmatrix} 6 & 1 \\ 3 & 2 \end{bmatrix} \begin{bmatrix} 4 & 3 \\ 1 & 2 \end{bmatrix} = \begin{bmatrix} 25 & 20 \\ 14 & 13 \end{bmatrix}$$

and

$$\det AB = 25 \cdot 13 - 20 \cdot 14 = 325 - 280 = 45$$

- On the other hand, since $\det A = 9$ and $\det B = 5$,

$$(\det A)(\det B) = 9 \cdot 5 = 45 = \det AB$$

Suggested Exercises

- 3.2.5
- 3.2.9

3.3. Cramer's Rule, Volume, and Linear Transformations

Acknowledgement

- This lecture note is based on the instructor's lecture notes (formatted as ppt files) provided by the publisher (Pearson Education) and the textbook authors (David Lay and others)
- The pdf conversion project for this chapter was possible thanks to the hard work by HyeonYeong Seo (ITM 19').
- In her junior year, she served as a vice president in the 10th ITM student council (year of 2021). In the Applied Probability Lab, she conducted market research projects for innovative service robot technologies.
- Upon her graduation, she has been with Microsoft Korea Inc. since 2023.

